

David Kreuter

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Senior Infrastructure & Networking Engineer | Distributed Systems, VPN, and API-Driven Automation

Enthusiastic technical professional with 10+ years of experience engineering complex networked systems at scale. Possesses deep expertise in IP management, VPN connectivity, DNS architecture, firewall automation, and distributed infrastructure spanning geo-redundant datacenters. Architected a global provisioning framework that manages 20,000+ virtual machines with integrated, API-driven workflows across VMware, Infoblox, Check Point, and Rubrik. Proven ability to diagnose and resolve complex networking issues, build for observability, and deliver secure, resilient infrastructure with a security-first mindset. Experienced with SQL databases, Python middleware development, and hybrid-cloud networking.

WORK EXPERIENCE

M&T Bank

03/2019 - Present

Automation Engineer (formerly Systems Engineer)

- **Operational Efficiency:** Developed an intelligent IP provisioning engine via Infoblox API integrations that automated IP address management, eliminated 1,000+ engineering hours annually, and resolved a multi-year infrastructure bottleneck in IP addressing and routing workflows.
- **Network Security Automation:** Automated Check Point firewall rule generation and DMZ provisioning; integrated security policy updates into the server build cycle to ensure day-zero network isolation, NAT configuration, and a hardened security posture across distributed environments.
- **Infrastructure Resilience:** Engineered and maintained a highly resilient distributed infrastructure architecture spanning three identical, geo-redundant datacenters with automated failover to ensure maximum uptime and disaster recovery readiness.
- **Architectural Scale:** Engineered a global server provisioning framework that deployed and managed 20,000+ virtual machines; integrated Rubrik and VMware APIs to automate backup and compute placement, built for observability, and ensured 100% RTO/RPO compliance at build time.
- **Platform Engineering & Middleware:** Designed and developed a custom Python orchestration application as middleware between ServiceNow and Ansible Automation Platform; implemented advanced retry logic, auto-remediation of deployment failures, and a pub/sub queue model for high-volume scaling in a distributed systems architecture.
- **Database as a Service (DBaaS):** Collaborated with Database Engineering to architect a high-performance on-premises SQL cluster, enabling a unified DBaaS offering that consolidated fragmented legacy databases and streamlined application delivery.
- **Cloud Abstraction & Redesign:** Refactored legacy on-premises automation workflows to be fully cloud-agnostic, enabling seamless deployment to Azure; designed the logic to be transparent to end users, maintaining a consistent interface while abstracting the underlying infrastructure provider.
- **Hybrid Cloud & GitOps:** Architected GitOps-driven CI/CD pipelines using Terraform to automate the deployment of virtual machines to Azure; implemented a pull-request-based workflow that standardized infrastructure changes through automated plan/apply cycles.
- **End-to-End Hardening & Provisioning:** Developed a zero-touch build workflow that delivers fully patched and hardened Linux (Ansible) and Windows (GPO) servers; automated agent installation, drive formatting, and optional hardened SQL deployments at scale with a security-first approach.
- **Technical Leadership:** Serve as a primary mentor for the Technology Development Program (TDP), setting enterprise-wide standards for naming conventions, Git branching strategies, and unit testing for infrastructure code.
- **Risk Mitigation:** Authored Active Directory lifecycle scripts with built-in 'read-only' dry-run modes; implemented least-privilege permissions and robust rollback protocols for stale object cleanup.
- **Cost Optimization:** Orchestrated the migration of 850 branch file servers within a strict 3-month window using custom PowerShell automation, avoiding \$400,000 in extended licensing fees.

OTG Management

07/2013 - 02/2019

Lead Network Engineer

- **Infrastructure Resilience:** Architected a highly redundant DNS solution that significantly reduced network outages and optimized global traffic-routing performance across distributed environments.
- **Cloud Connectivity:** Established secure, encrypted VPN tunnels to bridge on-premises networks with cloud resources, ensuring seamless hybrid-cloud operations and secure interconnectivity.
- **Security Architecture:** Engineered a secure reverse proxy solution to protect sensitive internal infrastructure, enabling customers' online ordering while mitigating external threat vectors.
- **Performance Engineering:** Diagnosed and resolved a mission-critical server hang by identifying a NUMA architecture mismatch; realigned vCPU/memory topology to eliminate cross-node latency and restore system stability.

Network Engineer

- **Network Administration:** Managed complex firewall rules, switch ACLs, and routing policies for a distributed network spanning 12 airports and 3 satellite offices, troubleshooting and resolving complex networking issues across diverse environments.
- **Proactive Monitoring:** Developed service-aware alerting and observability frameworks for business-critical systems, reducing mean time to resolution (MTTR) through proactive incident detection and response.

- **Cybersecurity Governance:** Directed enterprise-wide vulnerability scanning and remediation efforts for internal and external systems to maintain a hardened security posture.
- **Enterprise Mobility:** Successfully onboarded and managed 15,000 iPads via Meraki MDM, implementing automated security protocols including geofencing, remote-wipe triggers, and custom application deployment.

Systems Administrator

- **Operational Visibility:** Modernized infrastructure monitoring by designing custom Grafana dashboards to visualize real-time performance metrics, improving observability across networked systems.
- **Disaster Recovery:** Spearheaded the deployment of Datto backup and disaster recovery appliances to ensure 100% business continuity for critical operations.

SKILLS

Networking: VPN, DNS, IP Addressing & Management, BGP, F5 Load Balancing, High Availability, VRRP, Intrusion Detection, Vulnerability Scanning, Meraki MDM

Languages: Python, SQL, Bash, PowerShell, YAML, Javascript, PHP

Automation/Orchestration: REST API, Middleware Development, Terraform, GitOps, CI/CD Pipelines, Ansible Automation Platform, Grafana, ServiceNow, Gitlab, GraphQL, Unit Testing

Infrastructure: Infrastructure as Code (IaC), VMware vSphere, Azure Cloud Services, Check Point Firewalls, OS Hardening (CIS/STIG), DBaaS, Active Directory, Rubrik, Nutanix AHV, NUMA Tuning

EDUCATION

Bachelor Of Science in Computer Information Systems

12/2012

SUNY Fredonia

CERTIFICATIONS

Certified Meraki Network Operator

Meraki

VOLUNTEERING & LEADERSHIP

Habitat For Humanity

Volunteer

07/2025 - Present

Tift Nature Preserve

Volunteer

01/2025 - Present

716 Paws

Thermal drone pilot for search and rescue missions.

01/2022 - Present